

Labeled outerplanar graph expansions

Exact coefficients component refinements and safe inversion

Report 170

3 October 2026

Abstract

Let c_n and g_n count connected and arbitrary outerplanar simple graphs on the fixed label set $[n]$, respectively. These are the positive-index terms of OEIS A097998 and A098000. Starting with the classical block decomposition, we give an explicit finite algebraic rule for every fixed-order correction to

$$c_n \sim An! \rho^{-n} n^{-5/2}, \quad g_n \sim e^\nu An! \rho^{-n} n^{-5/2},$$

and evaluate the first four corrections. An exact component-marked coefficient identity leads to a uniform complex-parameter saddle proof with a Gaussian-integrable remainder and no logarithmic loss.

For the component count K_n we obtain second-order expansions of its mean, variance, and connectivity probability, and prove

$$d_{\text{TV}}(\mathcal{L}(K_n), \mathcal{L}(1 + \text{Poisson}(\nu + 5\tau/(2n)))) = O(n^{-2}).$$

A logarithmic expansion gives a Lambert- W starting point and a rounding-safe, two-ceiling enclosure for the generalized inverse. The companion supplies exact recurrences, algebraic certificates, literal small-graph enumeration, independent formal coefficient checks, and bounded regression tests.

The leading equivalents and limiting shifted-Poisson law are established results. General subcritical graph theory already implies fixed-order expansions. This report is an explicit specialization and reproducible refinement, without a worldwide-priority claim. Decimal evaluations are not certified intervals; inverse error constants and starting thresholds are existential rather than evaluated.

[write] **Status and trust boundary, added 6 October 2026, batch 109.** This is bundle Report 170 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a097998-outerplanar-graphs** (provenance, the sources as the write read them, what was checked, relation to the repository, the collected non-claims and the reading conventions are in Section 1.1). It is unrefereed and not formalized. Its author line reads “Report 170” and its PDF author field is empty: it names no person or tool. *What rests on what.* Proposition 2.1 (Lagrange inversion and one coefficient identity) and Theorem 3.1 (an exact contour integral, global dominance from the nonnegative coefficients of b , and a Taylor remainder kept inside the Gaussian integral) have conventional proofs; Theorem 6.1, Corollary 6.2 and Theorem 7.1 follow from Theorem 3.1 by Cauchy estimates, Stirling’s series and the mean value theorem. No proof uses a computation. The values $d_2(0)$, d_3 , d_4 and the field formula (5.5) are outputs of the finite rule (3.11), computed by the shipped programs; $d_1(v)$ and the shape of $d_2(v)$ are also derived by hand. Every decimal is a diagnostic. The leading equivalents and the shifted-Poisson limit are Bodirsky, Giménez, Kang and Noy’s (BGKN), the block decomposition is Bodirsky and Kang’s, and fixed-order expansions exist by the subcritical framework of Drmota, Fusy, Kang, Kraus and Rué. *A correction of the literature.* The amplitude $h \approx 0.018216$ printed in BGKN’s Theorem 5.1 is wrong: the true value is $e^\nu A = 0.0080960474\dots$, and the printed decimal is $\frac{9}{4}e^\nu A$ to its six places (Section 9); the write proves this in Remark 9.1, with an elementary

bound and an interval-arithmetic enclosure, and records the other versions, the introduction's exponent and decimals, and the same factor in BGKN's Theorem 7.1 in Remark 9.2. The write also adds Remark 1.1 (the OEIS entries quoted), Remarks 7.2 and 7.3 (the onset of monotonicity, and the inverse against the transseries volume), dated notes, and three questions in Section 10. No statement, proof or number of the source was changed.

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1 Objects conventions and scope

An outerplanar graph is an abstract simple graph that admits a planar embedding with all its vertices on the unbounded face. We count each edge set on $[n] = \{1, \dots, n\}$ once, regardless of the number of its embeddings. Neither plane maps nor unlabelled isomorphism classes are being counted.

For $n \geq 1$, put $c_n = \text{A097998}(n)$, and for $n \geq 0$ put $g_n = \text{A098000}(n)$ [1, 2]. The initial terms are

n	0	1	2	3	4	5
c_n in the EGF	0	1	1	4	37	602
g_n	1	1	2	8	63	893

The OEIS connected sequence conventionally prepends 1 at index zero. That convention is not used in the connected exponential generating function (EGF):

$$C(z) = \sum_{n \geq 1} c_n \frac{z^n}{n!}, \quad G(z) = \sum_{n \geq 0} g_n \frac{z^n}{n!} = e^{C(z)}. \quad (1.1)$$

Including the prepended term in C would incorrectly multiply G by e . A block means a 2-connected outerplanar graph on at least three vertices, or the single edge K_2 . There is no singleton block.

Bodirsky and Kang develop exact labeled enumeration and sampling [3]; Bodirsky, Giménez, Kang, and Noy (BGKN) establish the leading count equivalents and the limiting component law [4]. Kang's thesis also supplies the correct analytic amplitude relation [6]. The analytic framework for

labeled subcritical graph classes includes this family [7]. We retain these attributions throughout. The useful content here is the finite exact coefficient generator, four computed corrections, explicit finite-size component refinements, a direct complete remainder proof, and the integer-safe inverse construction.

Every asymptotic assertion below takes its order J or m fixed before n tends to infinity. No convergence of the resulting infinite formal series is asserted. Uniformity in a marker means uniformity on each fixed compact subset of $\mathbb{C}$, not for markers growing with n .

Remark 1.1 ([write] The OEIS entries, quoted, 6 October 2026). The entries were read on 6 October 2026 in the OEIS internal format; the quotations are verbatim.

- (a) A097998 [1] (revision #12, 12 February 2021) is named

Number of connected outerplanar graphs on n labeled nodes.

with offset 0 and data 1, 1, 1, 4, 37, 602, 14436, 458062, ...; its edit line reads “a(0)=1 prepended and terms a(17) and beyond from _Andrew Howroyd_, Feb 12 2021”, the prepended term that Section 1 excludes from C . Its only formula is “Recurrence known, see Bodirsky and Kang.”, and its PARI program (Howroyd) is

`seq(n)={Vec(serlaplace(1 + intformal(serreverse(x/exp((1 + 5*x -
sqrt(1 - 6*x + x^2 + O(x^n)))/8))/x))))}`

that is, T as the compositional inverse of $ue^{-b(u)}$ and $C = \int_0^z T(s) ds/s$: the rooted equation (2.3) with the block function (2.1), plus the prepended 1.

- (b) A098000 [2] (revision #14, 12 February 2021) is named “Number of outerplanar graphs on n labeled nodes.”, with offset 0, data 1, 1, 2, 8, 63, 893, 19714, ..., the formulas “Recurrence known, see Bodirsky and Kang.” and “Exponential transform of A097998. - _Andrew Howroyd_, Feb 12 2021”, and the same program with `exp(...)` in place of `1 + ....`
- (c) A097999 [10] (revision #42, 30 May 2026), “Number of 2-connected outerplanar graphs on n labeled nodes.”, offset 3: its eighteen printed terms are $(n-1)! [u^{n-1}]b(u)$ for $3 \leq n \leq 20$, by the write’s own computation from $S = 1 + uS + uS^2$, so (2.2) is the derivative of its exponential generating function plus u , the edge K_2 . It links the extended abstract [11] and Finch’s note [8].

Neither A097998 nor A098000 states an asymptotic formula or a constant. Both b-files (Howroyd, $0 \leq n \leq 200$), fetched on 6 October 2026, are byte-identical to the shipped fixtures (SHA-256 b150fb8f...56bb66 and ceb4c0e1...ac50b). With its own code the write recomputed every c_n , $1 \leq n \leq 200$, from (2.4) and every g_n , $0 \leq n \leq 200$, from $G = e^C$, in exact rational arithmetic: all 401 terms agree with the b-files. Nothing was submitted to the OEIS.

1.1 Provenance, sources and reading conventions

[write] **Added 6 October 2026, batch 109.** *Provenance.* This report is Report 170 of the session bundle that ProveIt’s docs/incoming intake processed as batch 109: the archive `Labeled_Outerplanar_Graphs_Expansions_and_Inverses_Source.zip` (553,249 bytes, SHA-256 bfdcc45dce2d...ee19e154abb6f45, 29 files, no wrapper directory), arrival commit 60f54ea06, placed in f7e9e5c2f. The text is the delivered

`Report170.tex` (921 lines, dated 3 October 2026), shipped as `article.tex`. The companion programs are shipped as `code/companion-*.py`, the root `build_pdf.py`, `make_zip.py` and `release_tools.py` under `code/`, the fixtures and recorded results as `data/companion-fixtures-*` and `data/companion-results-*`, `SOURCE_PROVENANCE.json` under `data/`, and the companion README as `companion-README.md` (the report README maps every name). Not shipped, and retrievable from the arrival commit: the delivered 15-page PDF, the checksum manifests `SHA256SUMS` (28 entries) and `companion/results/SHA256SUMS` (5 entries), both verified at placement, the byte-identical second copy `companion/release_tools.py`, and the delivered README, replaced by the report README. The package records no ProveIt commit and names no repository path; it carries no “prepared for private review” line and no personal data. It is a single-source report, so no merge choice was made. The write prefixed the 49 delivered labels with `lop:` and updated the 37 references to them; it added labels to the ten sections and to Section 5.1, and added this subsection, the notes marked [write], Remarks 1.1, 7.2, 7.3, 9.1 and 9.2, three questions in Section 10 and two bibliography entries. The added statements are the last of their sections, this subsection follows the last delivered text of Section 1, and the added displays are unnumbered, so every theorem, section and equation of the source keeps its delivered number.

The sources, as the write read them (6 October 2026).

- OEIS A097998, A098000 with their b-files, and A097999, in the internal format (Remark 1.1).
- BGKN, published version [4] (the author-hosted PDF the source names; text extracted, and the pages of Theorems 5.1 and 6.1 and of the introduction rendered): the introduction (pp. 2092–2093), Section 5 with Theorem 5.1 (p. 2102), Theorem 6.1 (p. 2103), Section 7 with Theorems 7.1 and 7.3 (p. 2104), the table of p. 2105, and, for the normalization, Theorems 3.6 and 3.7 (pp. 2099–2100).
- BGKN, arXiv:math/0512435 [5]: the abstract page lists only version 1 (19 December 2005); its Theorems 5.1, 6.1 and 7.1 and its introduction were read.
- Kang’s habilitation thesis [6]: Theorems 5.2.1 and 5.3.2 as stated in the overview (p. 9) and Section 5.2 (p. 55).
- The extended abstract of BGKN at EuroComb 2005 [11] (the copy cached at A097999): its Theorem 3.
- The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:def:core`, `p0:prop:factorial-core`, `p0:thm:core-reversion`, `p0:prop:operator-form`, `p0:def:three-inverses`, `p0:thm:staircase`, `plt:def:lw-monomial-log-datum` and `plt:thm:lw-template`, read for Remark 7.3.
- Not read by the write: Bodirsky and Kang [3], the subcritical framework [7], Finch’s note [8] and the unlabelled study [9]; the source’s account of them is reported, not confirmed.

What was checked at intake. The batch-109 dossier read the manuscript in full, found no error, and checked the proof of Section 4. Independently of the delivered code (its own Schröder and Lagrange route, 50 digits) it found the relative residual of c_n after the four corrections of (3.9) to be $1.71 \cdot 10^{-7}$ at $n = 60$ and $5.27 \cdot 10^{-9}$ at $n = 120$ (ratio $32.5 \approx 2^5$); it read BGKN’s Theorem 5.1 and introduction, and it reran the four computations of the companion on copies through a Windows driver (the delivered `build.py` needs POSIX `O_NOFOLLOW`): all four result files byte-identical. *What the write checked.* Every proof line by line, with no error found: Proposition 2.1 (the coefficient identity and $\psi' = 1 - ub'$); the contour formula (4.1) (the factors $\tau\rho^{-n}$, $e^{v\nu}$ and $e^{i\theta}$, and uQ' written through K'); the global bound (4.2); the derivative formula for $\partial_s^r S$, the degree count $r + 2r$ in (4.4) and the remainder (4.6); the arc estimates; the parity and degree arguments of Section 4; the finite sum (3.11) (the powers of i and κ_2 , and q even); $d_1(v)$ from $[t]M_v$, $[t^2]M_v$ by hand; the variance as $(\log P_n)''(1) + (\log P_n)'(1)$ and the connectivity coefficient $-h + da + a^2 - k$; $\ell_1, \dots, \ell_4$ against Stirling’s series; the injections of Section 7; and the test points, the choice $K_m = 8C_m$ and the mean value step of the proof of Theorem 7.1. With its own code (mpmath 1.3.0 at 80 digits, SymPy 1.14.0) it recomputed τ , ρ , ν , A , $e^\nu A$ and $\kappa_2, \dots, \kappa_{14}$ (with $\kappa_1 = 1$), the identity $\sqrt{1 - 6\tau + \tau^2} = \tau(3 - \tau)/(8 - 5\tau)$, the formulas (5.1) and (5.5), $d_1(v)$ and $d_2(v)$ at $v = 1, 2$, all eight entries of the table of Section 5 (each correctly rounded) from the finite sum, the constants of Section 6, and the residuals (notes at the ends of Sections 3, 6 and 8). The four companion computations were rerun from the shipped files (README, Route B): byte-identical results.

Finite and floating computations are observations, not certificates; the interval enclosure of Remark 9.1 is `mpmath` interval arithmetic.

Relation to the repository. Before batch 109 no file of the repository named A097998, A098000 or A097999, and no report counted outerplanar graphs. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The nearest reports are the graph-class enumerations in the same directory `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/`: `a005975-interval-graphs`, `a156808-circle-graphs` and `a123448-permutation-graphs`. They count unlabelled graphs, by different methods, and share no result with this report. The inverse of Section 7 is compared with the transseries volume in Remark 7.3.

[write] What is not claimed, collected from the source. No worldwide priority is claimed: the leading equivalents, the limiting shifted-Poisson law and the general machinery are prior, the existence of fixed-order expansions follows from the subcritical theory, and the source “does not announce a new leading constant”; its novelty statement is “a source coverage statement, not proof of worldwide novelty”. Every expansion is for a fixed order; no convergence of the formal series is asserted; uniformity in the marker holds on fixed compact sets only, not for markers growing with n . Decimal evaluations are not certified intervals. The inverse constants and starting thresholds are existential; Theorem 7.1 is not a certified finite-input algorithm, and the enclosure cannot be replaced by $N(X) = \lceil r_m(X) \rceil$. The formal singular route of Section 5.1 is an independent coefficient check, not a second analytic (Δ -domain) proof. The cause of the BGKN discrepancy is not established. Numerical tables and finite agreement are diagnostics; they do not prove convergence, bound the constants in $O(\cdot)$ or certify an inverse. The write adds: its own checks are floating, finite or interval computations; Remark 9.1 corrects a published decimal but finds no error in its exponent or in its radius; Remark 9.2(d) is an observation, not a theorem; Remark 7.3 claims no novelty for any inversion.

Reading conventions (letters the source reuses, with the tempting false readings; no symbol of the source was renamed; symbols of the transseries volume carry the subscript “vol” in Remark 7.3).

Symbol	Meaning here (where)	Not to be read as
a	$\sqrt{\kappa_2}$ in Section 4; $5\tau/2$ in (6.1) and after	each other
e	$d_2(0)$ in Section 6; elsewhere Euler’s number (e^ν , $e^{-\nu}$)	each other
h	the analytic factor of $R_3 = t^3h$ (Section 4); the v -coefficient of d_2 (Section 6); the inverse displacement (Section 7); BGKN’s amplitude, which is $e^\nu A$ (Remark 9.1)	each other; the volume’s h_{vol}
k, k_*	a summation index; the v^2 -coefficient $\frac{35}{8}\tau^2$ of d_2 ; $\log(A_*\sqrt{2\pi})$	each other
q, Q	the moment constant (6.1); $\ell + \sum km_k$ in (3.11); coefficients $q_{n,k}$ and the comparison PGF Q_n (proof of Theorem 6.1); the germ $Q(t)$ (3.5)	each other
S	the Schröder series (Section 2); the phase $S(s, x)$ (Section 4)	each other
A, A_v, A_*	the amplitude (3.3); $e^{v\psi}$ (proof of Proposition 2.1); A or $e^\nu A$ (Section 7)	BGKN’s dissection series $A(x, y)$
$E, \mathbb{E}$	$e^{nb(u)}$ (proof of Proposition 2.1); remainders E_n, E_m ; the Gaussian moment operation and expectation	each other
$K, \mathcal{K}, K_n, K_m$	the germ (3.4); a compact set of markers; the number of components; the constant of Theorem 7.1	each other
C, C_j	the connected EGF; the singular coefficients (5.7); C, C_r, C_m are also generic constants	each other
T, T_k, t_2	the rooted EGF; $[X^k]T$ (5.6); the constant (5.2)	t , the germ variable
W	$W(t)$ of Section 5.1; Lambert’s function in (7.3)	each other

Symbol	Meaning here (where)	Not to be read as
X, L	$\sqrt{1 - z/\rho}$ in Section 5.1; the threshold and $L = \log X$ in Section 7	each other; the volume's $X_{\text{vol}}, L_{\text{vol}}$
r, r_m, N	a real radius $r > 1$ (proof of Theorem 6.1); the model root, $r = r_m(X)$ (Section 7); $N = 2J + 2$ (Section 4) and $N(X)$	each other
R, R_3	$3 - 2\sqrt{2}$; the cubic remainder of K	Kang's R , which is ρ
$B, b; d, d_j; \kappa_j$	the block EGF and its derivative; $d_1(0)$ and the marker polynomials; $(D^j b)(\tau)$	the volume's $B_{\text{vol}}, d_{\text{vol}}, \kappa_{\text{vol}}$
τ, ρ, ν	the saddle, the radius, $\psi(\tau)$	BGKN's τ, ρ (but σ in its introduction), ξ ; Kang's $u_1, x_1 = R = \gamma^{-1}, C(R)$; the volume's ρ_{vol}
$A, e^\nu A$	connected and all-graph amplitudes	Kang's α_1, α_0 (also his c_1); BGKN's h is meant to be $e^\nu A$

2 Exact block and marked identities

2.1 Block generating function

Let B be the block EGF and write $b = B'$. The classical polygon-dissection normalization gives

$$b(u) = \frac{1 + 5u - \sqrt{1 - 6u + u^2}}{8}, \quad B(u) = \int_0^u b(t) dt. \quad (2.1)$$

The square root is the branch equal to 1 at zero. In particular,

$$b(u) = u + \frac{u^2}{2} + \frac{3u^3}{2} + \frac{11u^4}{2} + \frac{45u^5}{2} + O(u^6). \quad (2.2)$$

For a 2-connected outerplanar graph with $n \geq 3$, its outer Hamilton cycle is unique. There are $(n-1)!/2$ unoriented Hamilton cycles on the fixed labels, and for each cycle the allowable chords are polygon dissections. Thus the dissection count is multiplied by $(n-1)!/2$, not by the number of plane embeddings. The edge K_2 supplies the linear term u in b separately [3].

For an independent recurrence formulation, let

$$S(u) = 1 + uS(u) + uS(u)^2 = 1 + 2u + 6u^2 + 22u^3 + 90u^4 + \dots$$

The branch with constant term 1 is $S = (1 - u - \sqrt{1 - 6u + u^2})/(2u)$, and $b = u + u(S - 1)/4$. This also checks (2.2). For example, there are nine 2-connected outerplanar graphs on four labels, so the coefficient of u^3 in b is $9/3! = 3/2$.

2.2 Rooting and integration

Define the vertex-rooted connected EGF $T = zC'(z)$. The rooted block set decomposition and subsequent integration give

$$T = z \exp(b(T)), \quad C(z) = \psi(T(z)), \quad \psi(u) = u - ub(u) + B(u). \quad (2.3)$$

Indeed $\psi'(u) = 1 - ub'(u)$, and $dz/z = (1/u - b'(u)) du$ under $z = ue^{-b(u)}$. Both sides of $C = \psi(T)$ vanish at zero. Lagrange inversion applied to T therefore gives the exact connected coefficient identity

$$\frac{c_n}{n!} = \frac{1}{n^2} [u^{n-1}] e^{nb(u)}, \quad n \geq 1. \quad (2.4)$$

The component-marked polynomial is

$$f_n(v) = n! [z^n] e^{vC(z)}. \quad (2.5)$$

For $n \geq 1$, $f_n(1) = g_n$, $f_n(0) = 0$, and $f'_n(0) = c_n$. For a uniformly chosen graph counted by g_n , the probability generating function (PGF) of its number K_n of components is $f_n(v)/g_n$.

Proposition 2.1 (Exact marked cancellation). *For every $n \geq 1$ and every complex v ,*

$$\frac{f_n(v)}{n!} = \frac{v}{n^2} [u^{n-1}] e^{v\psi(u)} (1 + vu\psi'(u)) e^{nb(u)}. \quad (2.6)$$

Proof. Lagrange inversion initially gives $(v/n)[u^{n-1}] e^{v\psi(u)} \psi'(u) e^{nb(u)}$. Set $A_v(u) = e^{v\psi(u)}$ and $E(u) = e^{nb(u)}$. Coefficient differentiation implies

$$(n-1)[u^{n-1}] A_v E = [u^{n-1}] u(A'_v + nb' A_v) E.$$

Rearranging, and using $A'_v = v\psi' A_v$, gives

$$[u^{n-1}] A_v E (1 - ub') = \frac{1}{n} [u^{n-1}] A_v E (1 + vu\psi').$$

Since $\psi' = 1 - ub'$, this proves (2.6). $\square$

The cancellation removes an amplitude that vanishes at the saddle. Keeping its exact extra factor $1/n$ is essential for the unrooted exponent $n^{-5/2}$ and its amplitude. No asymptotic approximation has been used in these identities.

3 Saddle constants and exact coefficient generator

Let $R = 3 - 2\sqrt{2}$, the positive branch singularity of b . The nonnegative coefficients of b show that $ub'(u)$ is strictly increasing from zero to infinity on $(0, R)$. There is a unique $\tau \in (0, R)$ with

$$\tau b'(\tau) = 1. \quad (3.1)$$

Equivalently, it is the root in $(0.17076, 0.17077)$ of

$$3\tau^4 - 28\tau^3 + 70\tau^2 - 58\tau + 8 = 0 \quad (3.2)$$

that satisfies the unsquared equation (3.1). The unsquared condition fixes the radical branch; merely squaring an equation is not a root-selection rule.

Put

$$\rho = \tau e^{-b(\tau)}, \quad \nu = \psi(\tau), \quad D = u \frac{d}{du}, \quad \kappa_j = (D^j b)(\tau), \quad A = \frac{\tau}{\sqrt{2\pi\kappa_2}}. \quad (3.3)$$

Then $\kappa_1 = 1$ and $\kappa_2 > 0$. Useful approximate values are

$$\begin{aligned}\tau &= 0.170764986812629023408139791998\dots, \\ \rho &= 0.136593733696039583904892236367\dots, \\ \rho^{-1} &= 7.32098005480462334884554918493\dots, \\ \nu &= 0.148886742687358857557744764544\dots, \\ A &= 0.00697609450712314901934080444574\dots, \\ e^\nu A &= 0.00809604747608070499620425650041\dots\end{aligned}$$

These are numerical evaluations, not certified decimal intervals. The exact definitions are (3.1)–(3.3).

At the selected root,

$$\sqrt{1 - 6\tau + \tau^2} = \frac{\tau(3 - \tau)}{8 - 5\tau}.$$

Consequently all the κ_j belong to $\mathbb{Q}(\tau)$. The correction coefficients below also lie in this field. The leading constant A itself includes the universal Gaussian factor $1/\sqrt{2\pi}$; $A\sqrt{2\pi} = \tau/\sqrt{\kappa_2}$ is algebraic. We do not call A an algebraic number.

Define formal analytic germs at $t = 0$ by

$$K(t) = b(\tau e^t) = \sum_{j \geq 0} \frac{\kappa_j}{j!} t^j, \quad (3.4)$$

$$Q(t) = \psi(\tau e^t) - \nu, \quad Q'(t) = \tau e^t(1 - K'(t)), \quad Q(0) = 0, \quad (3.5)$$

$$M_v(t) = e^{t+vQ(t)}(1 + vQ'(t)). \quad (3.6)$$

Here $Q(t) = O(t^2)$ and $M_v(0) = 1$. Let $\mathbb{E}$ denote the formal Gaussian moment operation on polynomials in x , determined by $\mathbb{E}x^{2h} = (2h - 1)!!$ and $\mathbb{E}x^{2h+1} = 0$, with $(-1)!! = 1$.

Theorem 3.1 (All fixed orders with a complex marker). *For every fixed $J \geq 0$ and every compact $\mathcal{K} \subset \mathbb{C}$,*

$$\frac{f_n(v)}{v e^{\nu} A n! \rho^{-n} n^{-5/2}} = \sum_{j=0}^J \frac{d_j(v)}{n^j} + O_{\mathcal{K}, J}(n^{-J-1}), \quad (3.7)$$

uniformly for $v \in \mathcal{K}$, with the removable value used at $v = 0$. Here $d_0 = 1$, $d_j \in \mathbb{Q}(\tau)[v]$, $\deg_v d_j \leq j$, and the exact finite coefficient rule is

$$d_j(v) = [\epsilon^{2j}] \mathbb{E} \left[M_v \left(\frac{i\epsilon x}{\sqrt{\kappa_2}} \right) \exp \left\{ \sum_{k \geq 3} \frac{\kappa_k (ix)^k \epsilon^{k-2}}{k! \kappa_2^{k/2}} \right\} \right]. \quad (3.8)$$

Only $k \leq 2j + 2$ and amplitude powers at most $2j$ are needed. In particular,

$$c_n = A n! \rho^{-n} n^{-5/2} \left(\sum_{j=0}^J \frac{d_j(0)}{n^j} + O_J(n^{-J-1}) \right), \quad (3.9)$$

$$g_n = e^\nu A n! \rho^{-n} n^{-5/2} \left(\sum_{j=0}^J \frac{d_j(1)}{n^j} + O_J(n^{-J-1}) \right). \quad (3.10)$$

For readers preferring a formula without a formal expectation, put $a_\ell(v) = [t^\ell]M_v(t)$. Sum over $\ell \geq 0$ and nonnegative integers m_k , $3 \leq k \leq 2j+2$, satisfying

$$\ell + \sum_{k=3}^{2j+2} (k-2)m_k = 2j.$$

If $q = \ell + \sum km_k$, then

$$d_j(v) = \sum \frac{a_\ell(v)(-1)^{q/2}(q-1)!!}{\kappa_2^{q/2}} \prod_{k=3}^{2j+2} \frac{(\kappa_k/k!)^{m_k}}{m_k!}. \quad (3.11)$$

The constraint makes q even. This is a finite algebraic formula for every coefficient and makes no reference to fitted count data.

[write] The decimals of this section, 6 October 2026. Recomputed by the write (mpmath, 80 digits, own code; $\kappa_1 = 1$ to working precision). Of the six values printed above with "...", four are truncations (τ , ρ , ρ^{-1} , A) and two are rounded in the last printed digit: $\nu = 0.14888674268735885755774476454361\dots$ (the display ends $\dots 764544$) and $e^\nu A = 0.00809604747608070499620425650040521\dots$ (the display ends $\dots 650041$); both displayed values are correct to within one unit of their last digit. Mpmath interval arithmetic from the sign change of $ub'(u) - 1$ at the two ends of

$$\tau \in [0.17076498681262902340813979199814, 0.17076498681262902340813979199815]$$

encloses

$$e^\nu A \in [0.00809604747608070499620425650036, 0.00809604747608070499620425650046]$$

(Remark 9.1). The source's bracket $(0.17076, 0.17077)$ for τ is confirmed the same way ($ub'(u) - 1 = -0.00277\dots$ and $+0.00281\dots$ at its ends), and $\kappa_2 = 95.3658927155668778\dots$

4 Complete uniform saddle proof

4.1 Exact contour and the unique dominant point

Cauchy's formula on $u = \tau e^{i\theta}$ transforms Proposition 2.1 into the exact identity

$$f_n(v) = \frac{n!v\tau e^{v\nu}\rho^{-n}}{n^2} \frac{1}{2\pi} \int_{-\pi}^{\pi} M_v(i\theta) e^{n\Phi(\theta)} d\theta, \quad (4.1)$$

where

$$\Phi(\theta) = K(i\theta) - K(0) - i\theta.$$

Every power of n and τ in (4.1) is exact. The extra $e^{i\theta}$ from extracting $[u^{n-1}]$ is the factor e^t in M_v .

Write $b(u) = \sum_{k \geq 1} b_k u^k$. Since $b_k \geq 0$ and $b_1 = 1$,

$$\operatorname{Re} \Phi(\theta) = \sum_{k \geq 1} b_k \tau^k (\cos(k\theta) - 1) \leq \tau(\cos \theta - 1). \quad (4.2)$$

On $[-\pi, \pi]$, this is strictly negative except at zero. Thus the whole integration circle has one dominant saddle. On closed arcs away from zero there is a fixed strictly negative bound. Also

$$\Phi(\theta) = -\frac{\kappa_2 \theta^2}{2} + O(\theta^3),$$

so there are fixed $\eta, c > 0$ with $\operatorname{Re} \Phi(\theta) \leq -c\theta^2$ for $|\theta| \leq \eta$. The amplitude is uniformly bounded for v in each fixed compact set and $\theta \in [-\pi, \pi]$, because $\tau < R$ and its dependence on v is entire. These observations prove both global dominance and marker uniformity without invoking a Δ -domain for $T(z)$.

4.2 An integrable remainder

Let $a = \sqrt{\kappa_2}$ and $\epsilon = n^{-1/2}$. On the central arc use

$$\theta = \epsilon x/a, \quad |x| \leq \epsilon^{-1/5}.$$

The angular cutoff is $\epsilon^{4/5}/a$, a fixed multiple of $n^{-2/5}$. Set

$$R_3(t) = K(t) - K(0) - t - \kappa_2 t^2/2 = t^3 h(t),$$

where h is analytic in a fixed neighborhood of zero, and define

$$\begin{aligned} S(s, x) &= s^{-2} R_3(isx/a) = s(isx/a)^3 h(isx/a), \\ U_v(s, x) &= M_v(isx/a) e^{S(s, x)}. \end{aligned}$$

The value at $s = 0$ is understood by analytic continuation. After leading normalization, the central integral is

$$\frac{1}{\sqrt{2\pi}} \int_{|x| \leq \epsilon^{-1/5}} e^{-x^2/2} U_v(\epsilon, x) dx. \quad (4.3)$$

Fix J and put $N = 2J + 2$. For $0 \leq s \leq \epsilon$ on this arc, the arguments isx/a lie in a fixed smaller disk of analyticity once ϵ is small. Uniform derivative bounds give

$$|S(s, x)| \leq C\epsilon|x|^3 \leq C\epsilon^{2/5} \leq C, \quad |\partial_s^r S(s, x)| \leq C_r |x|^{r+2} \quad (1 \leq r \leq N).$$

For completeness, writing $w = ix/a$ shows the latter estimate directly:

$$\partial_s^r S = s w^{r+3} h^{(r)}(sw) + r w^{r+2} h^{(r-1)}(sw).$$

The factor $s|w|$ is uniformly bounded. Similarly, $|\partial_s^r M_v(isx/a)| \leq C_{\mathcal{K}, r} |x|^r$. The product rule and the finite Bell-polynomial formula for derivatives of an exponential now imply

$$|\partial_s^r U_v(s, x)| \leq C_{\mathcal{K}, r} (1 + |x|)^{3r}, \quad 0 \leq r \leq N. \quad (4.4)$$

Indeed, a product of derivatives of S whose derivative orders sum to r has degree at most $r + 2r = 3r$ in this bound; derivatives of M_v do not increase it. The factor e^S is bounded by e^C .

Taylor's theorem in the real parameter s , with its integral remainder, therefore gives

$$U_v(\epsilon, x) = \sum_{k=0}^{2J+1} p_k(v, x) \epsilon^k + E_n(v, x), \quad (4.5)$$

$$|E_n(v, x)| \leq C_{\mathcal{K}, J} \epsilon^{2J+2} (1 + |x|)^{6J+6}. \quad (4.6)$$

Keeping the polynomial inside the Gaussian integral yields

$$\int e^{-x^2/2} |E_n(v, x)| dx \leq C'_{\mathcal{K}, J} \epsilon^{2J+2}.$$

Thus there is no logarithmic loss. Taking the supremum of the polynomial on a growing interval before integrating would not give the required sharp estimate.

Each p_k has parity k in x : an amplitude monomial has equal x - and ϵ -degrees, while a phase monomial has degrees q and $q - 2$. All odd terms integrate to zero on the symmetric interval.

Extending each even polynomial integral to the real line incurs an error smaller than every fixed power of ϵ . The omitted intermediate arc contributes at most a constant times

$$\epsilon^{-1}e^{-c'\epsilon^{-2/5}},$$

and the distant arc contributes $O_{\mathcal{K}}(\epsilon^{-1}e^{-c''/\epsilon^2})$ after leading normalization. Both are smaller than every fixed power. Equations (4.3)–(4.6) prove (3.7), with (3.8) obtained by the finite Taylor multiplication.

4.3 Algebraic coefficients and marker degree

Since $Q' = \tau e^t(1 - K')$ and $Q(0) = 0$, all Taylor coefficients of Q lie in $\mathbb{Q}(\tau)$. Gaussian moments supply rational factors; only even total powers of x survive, so the square roots of κ_2 cancel. Hence $d_j(v) \in \mathbb{Q}(\tau)[v]$.

For $h \geq 1$, the coefficient of v^h in $e^{vQ}(1 + vQ')$ starts at degree at least $2h - 1$ in t , because $Q = O(t^2)$ and $Q' = O(t)$. Multiplication by e^t does not lower that degree. To contribute to ϵ^{2j} requires $2h - 1 \leq 2j$, which for integer h gives $h \leq j$. This proves the degree bound. The proof of Theorem 3.1 is complete.

5 Four corrections and independent coefficient checks

Set $d = d_1(0)$ and define

$$d = -\frac{1}{2\kappa_2} + \frac{\kappa_3}{2\kappa_2^2} + \frac{\kappa_4}{8\kappa_2^2} - \frac{5\kappa_3^2}{24\kappa_2^3}, \quad (5.1)$$

$$t_2 = \tau \left(\frac{1}{\kappa_2} - \frac{\kappa_3}{3\kappa_2^2} \right). \quad (5.2)$$

Then the first two complete marker polynomials are

$$d_1(v) = d + \frac{5\tau}{2}v, \quad (5.3)$$

$$d_2(v) = d_2(0) + \left(\frac{7}{2}\tau d - \frac{35}{8}t_2 \right)v + \frac{35}{8}\tau^2 v^2. \quad (5.4)$$

Here (3.8) at $j = 2, v = 0$ is a short exact definition of $d_2(0)$; no decimal is needed to specify the theorem. For example,

$$[t]M_v = 1 - v\tau\kappa_2, \quad [t^2]M_v = \frac{1}{2} - \frac{v\tau}{2}(5\kappa_2 + \kappa_3).$$

Substitution in (3.11) gives (5.3) directly. The analogous finite algebra gives (5.4).

j	Connected $d_j(0)$	All $d_j(1)$
1	2.28327991621488764390	2.71019238324646020242
2	5.05931216558214791991	7.37088006509066565629
3	12.62518674698345424780	22.53510992886156396083
4	37.15520265458457580487	78.66695831469578724957

The table uses approximate evaluations of exact algebraic quantities. For all four rows, the companion encodes the exact values in the basis $1, \tau, \tau^2, \tau^3$ of

$$\mathbb{Q}[t]/(3t^4 - 28t^3 + 70t^2 - 58t + 8),$$

with the real root selected by (3.1). As one readable example,

$$d = \frac{146018067}{47538652} - \frac{892831981}{178269945}\tau + \frac{834955523}{356539890}\tau^2 - \frac{63474793}{237693260}\tau^3. \quad (5.5)$$

The finite generator is a more compact exact presentation of higher orders than expanding every large rational coefficient in print.

5.1 An independent formal singular expansion

This subsection checks coefficients independently; the analytic proof is already complete. Put $X = \sqrt{1 - z/\rho}$ and $t = \log(T/\tau)$. Then

$$1 - X^2 = \exp(t - K(t) + K(0)), \quad W(t) = \frac{1 - \exp(t - K(t) + K(0))}{t^2}, \quad W(0) = \kappa_2/2.$$

The physical branch is $X = -t\sqrt{W(t)}$. Formal Lagrange inversion gives

$$T_k := [X^k]T = \frac{\tau(-1)^k}{k} [t^{k-1}]e^t W(t)^{-k/2}, \quad k \geq 1, \quad (5.6)$$

with $T_0 = \tau$. Termwise integration of $zC' = T$ gives

$$C_0 = \nu, \quad C_1 = 0, \quad C_j = -\frac{2}{j} \sum_{\substack{r \geq 0 \\ j-2-2r \geq 0}} T_{j-2-2r} \quad (j \geq 2). \quad (5.7)$$

In particular,

$$T_1 = -\tau\sqrt{2/\kappa_2}, \quad T_2 = t_2, \quad C_2 = -\tau, \quad C_3 = \frac{2}{3}\tau\sqrt{2/\kappa_2}, \quad C_4 = -\frac{\tau + t_2}{2}.$$

Since $\Gamma(-3/2) = 4\sqrt{\pi}/3$,

$$\frac{C_3}{\Gamma(-3/2)} = A, \quad \frac{[X^3]e^C}{\Gamma(-3/2)} = e^\nu A. \quad (5.8)$$

This makes the factorial and Gamma normalization transparent.

For noninteger α the exact coefficient identity is

$$[z^n](1 - z/\rho)^\alpha = \rho^{-n} \frac{\Gamma(n - \alpha)}{\Gamma(-\alpha)\Gamma(n + 1)}.$$

The fixed-order formal Gamma-ratio expansion follows from

$$\log \frac{\Gamma(n + a)}{\Gamma(n + b)} = (a - b) \log n + \sum_{k \geq 1} \frac{(-1)^{k+1} \{B_{k+1}(a) - B_{k+1}(b)\}}{k(k + 1)n^k}.$$

Here the infinite sum is used only after truncation to a fixed order. For an independently coded alternative, expand $b(\tau + y)$ in ordinary Taylor powers, invert $X^2 = 1 - (1 + y/\tau)e^{-(b(\tau+y)-b(\tau))}$, and

use the central-binomial expansion to handle half-integer Gamma ratios. The companion implements this second route without Euler derivative or Gaussian-moment inputs.

For example, $d = 15/8 - (5/2)C_5/C_3$. After division by $ve^{v\nu}$, the X^5 coefficient of e^{vC} is $C_5 + vC_2C_3$, and its X^7 coefficient is

$$C_7 + v(C_2C_5 + C_3C_4) + \frac{v^2}{2}C_2^2C_3.$$

The Gamma ratios $-5/2$ and $35/4$ then recover (5.3)–(5.4). To promote these formal local calculations to a separate analytic coefficient proof one must supply suitable Δ -domain continuation and transfer remainder hypotheses, as in the established subcritical theory [7]. That additional route is not silently assumed here.

6 Component law moments and connectivity

Put

$$a = \frac{5\tau}{2}, \quad q = \tau d - \frac{35}{8}t_2 + \frac{5}{2}\tau^2. \quad (6.1)$$

Dividing the marker expansion by its value at 1 gives, uniformly on fixed compact sets,

$$P_n(v) := \mathbb{E}v^{K_n} = ve^{\nu(v-1)} \left(1 + \frac{a(v-1)}{n} + O(n^{-2}) \right). \quad (6.2)$$

In particular, the classical limit is $1 + \text{Poisson}(\nu)$ [4]. The next statement strengthens the approximation at finite n .

Theorem 6.1 (Total variation refinement). *For $\lambda_n = \nu + 5\tau/(2n)$,*

$$d_{\text{TV}}(\mathcal{L}(K_n), \mathcal{L}(1 + \text{Poisson}(\lambda_n))) = O(n^{-2}).$$

Proof. The parameter is positive. Fix a real $r > 1$. The comparison PGF is $Q_n(v) = v \exp(\lambda_n(v-1))$. Equation (6.2), proved complex-uniformly on the circle $|v| = r$, gives

$$\sup_{|v|=r} |P_n(v) - Q_n(v)| \leq C_r n^{-2}$$

for all sufficiently large n . Write $p_{n,k} = [v^k]P_n$ and $q_{n,k} = [v^k]Q_n$. Cauchy's coefficient estimate gives

$$|p_{n,k} - q_{n,k}| \leq C_r n^{-2} r^{-k} \quad (k \geq 0).$$

Both constant coefficients are zero for $n \geq 1$. Therefore the entire coefficient sum, including the Poisson tail above n , satisfies

$$\frac{1}{2} \sum_{k \geq 1} |p_{n,k} - q_{n,k}| \leq \frac{C_r}{2n^2} \sum_{k \geq 1} r^{-k} = \frac{C_r}{2(r-1)n^2}.$$

This is exactly the total variation distance. □

To obtain second-order moments, write $d_2(v) = e + hv + kv^2$, where

$$e = d_2(0), \quad h = \frac{7}{2}\tau d - \frac{35}{8}t_2, \quad k = \frac{35}{8}\tau^2.$$

On a fixed sufficiently small disk about $v = 1$, the normalized expansion is nonzero for all sufficiently large n . Taking its analytic logarithm gives

$$\begin{aligned} \log P_n(v) &= \log v + \nu(v-1) + \frac{a(v-1)}{n} \\ &\quad + \frac{q(v-1) + (5\tau^2/4)(v-1)^2}{n^2} + O(n^{-3}). \end{aligned} \quad (6.3)$$

Indeed the second-order polynomial before simplification is $(h-da)(v-1) + (k-a^2/2)(v^2-1)$ and $k-a^2/2 = 5\tau^2/4$. The error is analytic and uniformly $O(n^{-3})$ on a slightly larger disk, so Cauchy estimates justify differentiating it.

Corollary 6.2 (Moments and connectivity). *The component mean and variance satisfy*

$$\mathbb{E}K_n = 1 + \nu + \frac{a}{n} + \frac{q}{n^2} + O(n^{-3}), \quad (6.4)$$

$$\text{Var } K_n = \nu + \frac{a}{n} + \frac{q + 5\tau^2/2}{n^2} + O(n^{-3}). \quad (6.5)$$

The probability of connectivity satisfies

$$\frac{c_n}{g_n} = e^{-\nu} \left[1 - \frac{5\tau}{2n} + \frac{-\tau d + (35/8)t_2 + (15/8)\tau^2}{n^2} + O(n^{-3}) \right]. \quad (6.6)$$

Proof. The first derivative of $\log P_n$ at 1 is the mean; its second derivative plus its first derivative is the variance. Apply these identities to (6.3). Dividing (3.9) by (3.10) gives the second coefficient $-h + da + a^2 - k = -\tau d + (35/8)t_2 + (15/8)\tau^2$ in (6.6). $\square$

Numerically,

$$\begin{aligned} a &= 0.42691246703157255852034948\dots, \\ q &= 1.28213096120148270341548384\dots, \\ q + 5\tau^2/2 &= 1.35503266300427611409382198\dots, \\ -\tau d + (35/8)t_2 + (15/8)\tau^2 &= -1.15455298304659423472839209\dots, \\ e^{-\nu} &= 0.861666699427543\dots \end{aligned}$$

In particular, connectivity approaches its limiting probability from below to first order. All higher fixed-order component and connectivity corrections can be obtained by algebraic operations on the same marked generator.

[write] **The decimals of this section, 6 October 2026.** Recomputed by the write from (5.1), (5.2) and (6.1), and the connectivity coefficient also as the ratio of the two expansions of Theorem 3.1 ($d_2(0) - d_2(1) - d_1(1)(d_1(0) - d_1(1))$), agreeing to 10^{-75}). All five displayed values are rounded in the last printed digit, not truncated: $a = 0.4269124670315725585203494799\dots$, $q = 1.2821309612014827034154838364\dots$, $q + 5\tau^2/2 = 1.3550326630042761140938219799\dots$, the connectivity coefficient $-1.1545529830465942347283920852\dots$ and $e^{-\nu} = 0.86166669942754276\dots$; each display is correct to within one unit of its last digit. BGKN's Theorem 6.1 prints 0.14889 and 0.86166 for ν and $e^{-\nu}$, and Kang's Theorem 5.3.2 the connectivity limit 0.861666 (Remark 9.2).

7 Logarithmic expansion and a safe threshold inverse

Take either $a_n = c_n$ or $a_n = g_n$. Write $A_* = A$ or $e^\nu A$, respectively, and $e_j = d_j(0)$ or $d_j(1)$. Combining Theorem 3.1 with Stirling's expansion gives, for every fixed integer $m \geq 0$,

$$\log a_n = F_m(n) + O(n^{-m-1}), \quad (7.1)$$

where

$$F_m(x) = x \left(\log \frac{x}{\rho} - 1 \right) - 2 \log x + \log(A_* \sqrt{2\pi}) + \sum_{j=1}^m \frac{\ell_j}{x^j}. \quad (7.2)$$

The ℓ_j are obtained by adding the Stirling series to the formal logarithm of $1 + \sum e_j x^{-j}$. In particular,

$$\begin{aligned} \ell_1 &= e_1 + \frac{1}{12}, \\ \ell_2 &= e_2 - \frac{1}{2}e_1^2, \\ \ell_3 &= e_3 - e_1e_2 + \frac{1}{3}e_1^3 - \frac{1}{360}, \\ \ell_4 &= e_4 - e_1e_3 - \frac{1}{2}e_2^2 + e_1^2e_2 - \frac{1}{4}e_1^4. \end{aligned}$$

Thus every $\ell_j \in \mathbb{Q}(\tau)$, for either sequence.

Both sequences are eventually strictly increasing. Adding the new label as an isolated vertex injects the all-graph class on $[n]$ into that on $[n+1]$, with other graphs outside its image. Attaching the new label as a leaf to vertex 1 similarly injects the connected class; for $n \geq 2$ a tree with the new vertex of degree two is outside the image. Define

$$N(X) = \min\{n \geq 1 : a_n \geq X\}$$

for sufficiently large real X .

Let $L = \log X$. A convenient explicit starting point for a continuous inverse is

$$x_0 = \frac{L}{W(L/(e\rho))}, \quad (7.3)$$

where W is the positive real Lambert function. It solves $x_0(\log(x_0/\rho) - 1) = L$. For all large L , F_m is strictly increasing in the relevant range and has a unique large root r_m of $F_m(r_m) = L$. Since the lower-order residual at x_0 is $O(\log x_0)$ and the derivative is asymptotic to $\log x_0$, $r_m = x_0 + O(1)$. One bounded correction is also explicit. Set $k_* = \log(A_* \sqrt{2\pi})$ and $\Lambda_0 = \log(x_0/\rho)$. For every fixed $m \geq 0$,

$$r_m = x_0 + \frac{2 \log x_0 - k_*}{\Lambda_0} + O\left(\frac{1}{x_0 \Lambda_0}\right). \quad (7.4)$$

To verify this, the displayed displacement h is bounded and tends to 2. Expanding $F_m(x_0 + h) - L$ gives $\Lambda_0 h - 2 \log x_0 + k_* + O(x_0^{-1}) = O(x_0^{-1})$. The curvature of the leading term, the change in $-2 \log x$, and any retained inverse-power terms are all $O(x_0^{-1})$. Division by $F'_m(x) \sim \Lambda_0$ gives (7.4). Thus $r_m - x_0 \rightarrow 2$; this is a statement about the continuous comparison root.

Newton iteration uses the explicit derivative

$$F'_m(x) = \log(x/\rho) - \frac{2}{x} - \sum_{j=1}^m \frac{j \ell_j}{x^{j+1}}. \quad (7.5)$$

Theorem 7.1 (Integer-safe inverse). *For each fixed $m \geq 0$ there is $K_m > 0$ such that, for all sufficiently large X , with $r = r_m(X)$ and*

$$\delta = \frac{K_m r^{-m-1}}{\log r},$$

one has

$$\lceil r - \delta \rceil \leq N(X) \leq \lceil r + \delta \rceil. \quad (7.6)$$

Proof. Write the error in (7.1) as $E_m(n)$, with $|E_m(n)| \leq C_m n^{-m-1}$ beyond a fixed integer threshold. Increase that threshold so that for $|x - r| \leq 2$,

$$F'_m(x) \geq \frac{1}{2} \log r, \quad |E_m(n)| \leq 2C_m r^{-m-1} \quad (n \in \mathbb{N}, |n - r| \leq 2).$$

Choose $K_m = 8C_m$, or any larger positive constant; enlarge C_m to be positive if necessary. Eventually $\delta < 1$. The explicit integer test points are

$$n_- := \lceil r - \delta \rceil - 1, \quad n_+ := \lceil r + \delta \rceil.$$

Both lie within distance two of r , with $n_- < r - \delta$ and $n_+ \geq r + \delta$. The mean value theorem gives

$$F_m(n_-) < L - 4C_m r^{-m-1}, \quad F_m(n_+) \geq L + 4C_m r^{-m-1}.$$

Allowing for E_m shows $a_{n_-} < X$ and $a_{n_+} \geq X$. Eventual monotonicity, together with X exceeding the finitely many earlier values, proves $n_- + 1 \leq N(X) \leq n_+$, which is (7.6). $\square$

The two ceilings agree except in a shrinking neighborhood of an integer. One cannot replace the enclosure by an unconditional formula $N(X) = \lceil r_m(X) \rceil$. Exact-threshold inputs $X = a_n$ are particularly sensitive: even a tiny positive error $r_m(X) - n$ makes the latter ceiling wrong. The companion can exhibit this numerically, but such diagnostics are not proofs of a remainder constant or of a finite-input enclosure. Theorem 7.1 is an asymptotic existence result. A certified finite-input algorithm would additionally require evaluated error constants, a starting threshold, and interval evaluation of the root.

Remark 7.2 ([write] The onset of monotonicity, 6 October 2026). The injections above give $g_{n+1} > g_n$ for every $n \geq 1$ and $c_{n+1} > c_n$ for every $n \geq 2$, while $g_0 = g_1 = 1$ and $c_1 = c_2 = 1$. *Proof.* For $n \geq 1$, adding $n + 1$ as an isolated vertex is injective (delete it again) and keeps outerplanarity (draw the new vertex in the outer face); the graph on $[n + 1]$ whose only edge is $\{1, n + 1\}$ is outerplanar and outside the image. For $n \geq 2$, adding $n + 1$ with the single edge $\{1, n + 1\}$ is injective and keeps connectivity and outerplanarity (vertex 1 lies on the outer face, and the pendant edge can be drawn there); every image has $n + 1$ of degree one, while the tree with edges $\{1, n + 1\}$, $\{n + 1, 2\}$ and $\{2, j\}$ for $3 \leq j \leq n$ (a path $1, n + 1, 2$ when $n = 2$) is connected, outerplanar like every tree, and has $n + 1$ of degree two. The equalities are the data of Remark 1.1. *Consequence.* For every $X > 1$, $N(X) = \min\{n \geq n_1 : a_n \geq X\}$ with $n_1 = 1$ for g and $n_1 = 2$ for c , a minimum over a strictly increasing sequence; the phrase “ X exceeding the finitely many earlier values” in the proof of Theorem 7.1 is then automatic, and only the threshold of the error bound remains existential.

Remark 7.3 ([write] The inverse and the transseries volume, 6 October 2026). Compared, statement by statement, with `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`; symbols of the volume carry the subscript “vol”.

- (a) *Instance.* By Remark 7.2, for $X > 1$ the threshold $N(X)$ is the integer staircase N_* of `p0: def:three-inverses` with $A_n = a_n$ and $n_1 = 1$ (for g) or 2 (for c), so `p0:thm:staircase(1)` gives $N(X) = \lceil \nu_{\mathcal{A}}(X) \rceil$ for every admissible interpolation $\mathcal{A}$. The source does not use this.
- (b) *Theorem 7.1: an analogue as stated and proved; also a consequence of `p0:thm:staircase(1)` once an interpolation is supplied.* The source proves the bracket directly, by the mean value theorem at two integer test points, about the root r_m of the model F_m , which does not interpolate the counts ($F_m(n) \neq \log a_n$ in general); its conclusion has the form of `p0:thm:staircase(2)`. A second route (added by the write): let $\tilde{E}$ be the piecewise-linear interpolation of the integer errors $E_m(n) = \log a_n - F_m(n)$ and put $\log \mathcal{A} = F_m + \tilde{E}$. On each $[n, n+1]$, $|\tilde{E}'| = |E_m(n+1) - E_m(n)| \leq 2C_m n^{-m-1}$ while $F'_m(x) \sim \log x$, so there is n'_1 beyond which $\mathcal{A}$ is continuous, strictly increasing and equal to a_n at the integers: an admissible interpolation. Let $\nu = \nu_{\mathcal{A}}(X)$, so that $F_m(\nu) - F_m(r) = -\tilde{E}(\nu)$. Since $|\tilde{E}(x)| \leq \max(|E_m(\lfloor x \rfloor)|, |E_m(\lceil x \rceil)|) \rightarrow 0$ and F'_m is eventually at least 1 on the segment between ν and r , eventually $|\nu - r| \leq 1$; then, with the threshold of the source's proof, $|\tilde{E}(\nu)| \leq 2C_m r^{-m-1}$ and $F'_m \geq \frac{1}{2} \log r$ on that segment, so $|\nu - r| \leq 4C_m r^{-m-1} / \log r \leq \delta$ (the source's $K_m = 8C_m$). Now `p0:thm:staircase(1)` and the monotonicity of the ceiling give $\lceil r - \delta \rceil \leq N(X) = \lceil \nu \rceil \leq \lceil r + \delta \rceil$ for all large X , which is (7.6).
- (c) *Instance.* The Lambert start (7.3) is `p0:prop:factorial-core` with $\kappa_{\text{vol}} = 1$ and $d_{\text{vol}} = -(1 + \log \rho)$: there $v = W_0(Le^{d_{\text{vol}}}) = W_0(L/(e\rho))$ and $X_{\text{vol}} = e^{v-d_{\text{vol}}} = e\rho e^v = L/v$, which is x_0 . Its large- L expansion is a formal instance of `plt:thm:lw-template`: with $\xi_{\text{vol}} = \log L$ and $Z_{\text{vol}} = \log x_0$ the equation $x_0(\log(x_0/\rho) - 1) = L$ reads $Z + \log Z + \log(1 - (1 + \log \rho)/Z) = \xi$, data $(\rho_{\text{vol}}, \mu_{\text{vol}}, \beta_{\text{vol}}, B_{\text{vol}}) = (1, 1, 0, \log(1 - (1 + \log \rho)t))$, $t = 1/Z$, over $\mathbb{R}$; it returns $x_0 = (L/\log L) \exp(\sum_{n \geq 1} q_n (\log \log L) (\log L)^{-n})$.
- (d) *Formal instance after a change of variable.* Put $x = x_0(1 + E)$, $t = 1/x_0$ and $\Lambda_0 = \log(x_0/\rho)$. Using $x_0(\Lambda_0 - 1) = L$, the equation $F_m(x) = L$ divided by x_0 becomes

$$\Lambda_0 E + h_{\text{vol}}(E) + f_{\text{vol}}(t, E) = 0, \quad h_{\text{vol}}(u) = (1 + u) \log(1 + u) - u,$$

$$f_{\text{vol}}(t, u) = t(k_* - 2 \log x_0 - 2 \log(1 + u)) + \sum_{j=1}^m \ell_j t^{j+1} (1 + u)^{-j},$$

which is `p0:thm:core-reversion` with $\Lambda_{\text{vol}} = \Lambda_0$, in a coefficient ring containing Λ_0^{-1} and $\log x_0 = \Lambda_0 + \log \rho$. Its first coefficient $e_1 = (2 \log x_0 - k_*)/\Lambda_0$ gives $x - x_0 = x_0 e_1 t + \dots = e_1 + \dots$, the displacement displayed in (7.4). Up to the terms $\ell_j x_0^{-j}/\Lambda_0$, it is also the $m = 1$ term of `p0:prop:operator-form`, since the core slope of $x(\log(x/\rho) - 1)$ is exactly Λ_0 . The instance is formal only; the remainder $O(1/(x_0 \Lambda_0))$ is the source's own analytic estimate.

- (e) *Not shown to be an instance of `plt:thm:lw-template`.* The full model equation $F_m(x) = L$ is not of monomial-logarithmic type in the two charts the write tested: with $Z_{\text{vol}} = x$ the balance $x \log x$ is not $\rho_{\text{vol}} Z + \mu_{\text{vol}} \log Z$ with constant $\rho_{\text{vol}}, \mu_{\text{vol}}$; with $\xi_{\text{vol}} = \log L$, $Z_{\text{vol}} = \log x$, the terms $-2 \log x + k_* + \sum_j \ell_j x^{-j}$ contribute Ze^{-Z} and e^{-Z} to the equation, which are not in $t \mathbb{R}[\log Z][[t]]$, $t = 1/Z$ (hypothesis (H3) fails). The write does not claim that no other change of variables makes it one; only the leading balance (c) is shown to be an instance.
- (f) The reversion (5.6) of Section 5.1 is ordinary power-series reversion. Formally it is the case $\mu_{\text{vol}} = 0$ of `plt:thm:lw-template`: with $\xi_{\text{vol}} = -1/X$ and $Z_{\text{vol}} = 1/t$ one has $-1/X = Z W(1/Z)^{-1/2} = \rho_{\text{vol}} Z + \beta_{\text{vol}} + B_{\text{vol}}(Z)$, $\rho_{\text{vol}} = \sqrt{2/\kappa_2}$, $B_{\text{vol}} \in t \mathbb{R}[[t]]$. The source uses Lagrange inversion directly and needs nothing from the volume.

No novelty is claimed for any of these inversions.

8 Reproducibility and the meaning of the checks

The companion is self-contained after its documented Python dependencies are installed. No network access is needed for its tests. It contains frozen OEIS b-files with source URLs, exact coefficient fixtures, portable programs, bounded commands, and deterministic release construction. The report and code use the same definitions, while several independent routes check their implementation.

The checks are deliberately separated by what they establish:

1. Exact block coefficients are computed from the square-root identity and independently from $S = 1 + uS + uS^2$. Integer Bell recurrences implement (2.4), and the exponential formula computes g_n . All 200 positive-index terms of each included OEIS b-file agree. The connected index-zero convention is checked separately.
2. Literal enumeration visits each simple edge set through $n = 5$. It accepts an edge set if some cyclic ordering of all vertices has no alternating-endpoint pair of edges, and tests connectivity directly. This is the circular straight-line characterization of outerplanarity. An edge set is counted only once, not once per successful ordering. It reproduces $c_1, \dots, c_5 = 1, 1, 4, 37, 602$ and $g_1, \dots, g_5 = 1, 2, 8, 63, 893$.
3. Exact arithmetic in $\mathbb{Q}(\tau)$ implements the finite Gaussian rule through four corrections and checks the marked identities (5.3)–(5.4), as well as the moment and connectivity algebra. Stored rational-basis certificates are compared exactly.
4. A separately coded ordinary-Taylor implicit inversion and central-binomial transfer checks the coefficient values at high precision. This is independent formal coefficient verification; it is not a second analytic Δ -domain proof.
5. Frozen decimal fixtures check numerical regression. Explicit failure predicates remain active under `python -O`. Generated results are compared between normal and optimized runs, and manifests check file integrity.

These finite checks do not prove asymptotic convergence, bound the constants hidden in $O(\cdot)$, or certify a finite-input inverse. Count-ratio, moment, and inverse residual tables, when produced, are diagnostics only. The analytic theorem rests on the exact identities and the uniform contour proof. High working precision and agreement between implementations do not turn approximate decimals into rigorous intervals.

The precise commands, dependency versions, source URLs, fixture hashes, and generated files are described in the companion’s README and source provenance file. Output directories must be new; an existing output is rejected rather than silently overwritten. The release manifest records the bytes actually delivered.

[write] Reruns in the repository, 6 October 2026. The companion’s README, its provenance file and the release manifest are shipped as `companion-README.md`, `data/companion-fixtures-provenance.json` and `data/companion-results-manifest.json`; the checksum manifests are not shipped, and the report README gives the retrieval of the delivered package and a route that rebuilds the companion’s layout from the shipped files. The delivered `build.py` and `tests.py` need POSIX `O_NOFOLLOW` and do not run on Windows; the four computations they drive were rerun by the dossier and again by the write (Route B,

about 11 s): all four result files byte-identical to the shipped ones, and all 15 input hashes of the shipped manifest equal to the shipped files. The residuals behind the corroboration in Remark 9.1, multiplied by n^5 , are 138.3, 133.3, 131.1, 130.3 for c_n and 333.1, 321.0, 315.8, 313.9 for g_n at $n = 30, 60, 120, 200$: consistent with a remainder of order n^{-5} in (3.9)–(3.10) at $J = 4$, and diagnostics only.

9 Source normalization and attribution

The leading outerplanar theorem in published BGKN [4, Theorem 5.1, pp. 2102–2103] uses the same unrooted labeled normalization as (3.10). It prints the all-graph amplitude as 0.018216. Equations (3.3) and (5.8) give instead

$$e^\nu A = 0.008096047476080704996204256500405217 \dots,$$

while

$$\frac{9}{4}e^\nu A = 0.018216106821181586241459577125911738 \dots$$

Thus the printed amplitude agrees, to its printed precision, with 9/4 times the value obtained from the defining generating functions. No cause of that discrepancy is established here.

Version distinctions matter. The earlier full preprint [5] prints 0.017657, a different decimal. The published component theorem already gives the compatible approximate parameter 0.14889 and connectivity probability 0.86166; its introductory decimals 0.14840 and 0.86208 are stale. The published introduction also retains an inconsistent $n^{-3/2}$ outerplanar exponent, while its main theorem has $n^{-5/2}$. These introductory issues do not alter the correct exponent in the theorem.

Kang’s thesis [6, p. 55] already gives the correct analytic connected-amplitude expression and the relation $\alpha_0 = \alpha_1 \exp(C(R))$. Its printed all-graph decimal 0.008095 is inaccurate, not just a lower-precision rounding of 0.008096047... The correct analytic normalization was therefore already present in the source set. This report resolves numerical normalization reproducibly; it does not announce a new leading constant.

The bounded source pass included the published and preprint BGKN versions, Kang’s thesis, the labeled section of the Bodirsky–Kang sampling paper, the subcritical graph-class framework, and Finch’s outerplanar growth-constant note [8]. Finch’s block convention excludes K_2 from his block function and compensates with an extra linear term in the rooted equation; combining them yields (2.1). Unlabelled outerplanar graph work [9] concerns different sequences and constants.

No explicit first-four correction table, two-order component-moment formula, or threshold inverse for these two labeled sequences was found in the inspected source portions and bounded searches. This is a source coverage statement, not proof of worldwide novelty. The leading theorem, the shifted-Poisson limit, and the general machinery remain prior work.

Remark 9.1 ([write] BGKN’s Theorem 5.1 corrected, with proof, 6 October 2026). The published Theorem 5.1 of [4] (p. 2102, read on the rendered page) states:

The number h_n of outerplanar graphs is asymptotically $h_n \sim h \cdot n^{-5/2} \cdot \rho^{-n} n!$ where $\rho \approx 0.13659$ and $h \approx 0.018216$.

Its h_n is g_n here, and its ρ is ρ here: on the same page BGKN’s $B'(x, 1)$ is b of (2.1), its $\Psi_1(u) = ue^{-B'(u, 1)}$, its $\tau \approx 0.17076$ is a root of (3.2), and $\rho = \Psi_1(\tau)$. *Claim.* The limit $\lim_n g_n / (n^{-5/2} \rho^{-n} n!)$ exists and equals $e^\nu A = 0.0080960474760807049962 \dots < 0.0084$. So the exponent $-5/2$ and the

radius are right, and the printed amplitude $h \approx 0.018216$ is wrong; it agrees, to its six printed decimals, with $\frac{9}{4}e^\nu A = 0.0182161068211815862414\dots$ (and $0.018216/(e^\nu A) = 2.24998\dots$).

Proof. (i) Theorem 3.1 with $J = 0$ and $v = 1$, that is (3.10), gives $g_n = e^\nu A n! \rho^{-n} n^{-5/2} (1 + O(n^{-1}))$; so the limit exists, equals $e^\nu A$, and any h with $g_n \sim h n^{-5/2} \rho^{-n} n!$ equals $e^\nu A$. (ii) An elementary bound. Differentiating $b'(u) = (5 - (u - 3)(1 - 6u + u^2)^{-1/2})/8$ gives $b''(u) = (1 - 6u + u^2)^{-3/2}$, so by (3.1) $\kappa_2 = \tau b'(\tau) + \tau^2 b''(\tau) = 1 + \tau^2(1 - 6\tau + \tau^2)^{-3/2}$. The root lies in $(0.17076, 0.17077)$ (Section 3; the sign change is confirmed in the note there). On this interval $1 - 6u + u^2$ is decreasing and lies between 0.004542 and 0.004600, so $(1 - 6\tau + \tau^2)^{3/2} < 0.000312$ and $\kappa_2 > 1 + 0.17076^2/0.000312 > 94.4$; hence $A = \tau/\sqrt{2\pi\kappa_2} < 0.17077/\sqrt{2\pi \cdot 94.4} < 0.007012$. Since b has nonnegative coefficients, it is nonnegative and increasing on $[0, \tau]$, so $0 \leq B(\tau) \leq \tau b(\tau)$ and $\nu = \psi(\tau) = \tau - \tau b(\tau) + B(\tau) \leq \tau < 0.17077$; thus $e^\nu < 1.1863$ and $e^\nu A < 0.00832 < 0.0084$. As $0.018216 > 2 \cdot 0.0084$, the printed value exceeds the true amplitude by more than a factor of two. (iii) An enclosure. On $[0, R]$ the function $F(u) = \frac{u-3}{2} \sqrt{1 - 6u + u^2} - 4 \log(3 - u - \sqrt{1 - 6u + u^2})$ is an antiderivative of $\sqrt{1 - 6u + u^2}$ (differentiate), so $B(\tau) = (\tau + \frac{5}{2}\tau^2)/8 - (F(\tau) - F(0))/8$, and ν , κ_2 and A are elementary in τ . Mpmath interval arithmetic at 40 digits, from the enclosure of τ in the note at the end of Section 3, gives

$$e^\nu A \in [0.00809604747608070499620425650036, 0.00809604747608070499620425650046],$$

$$\frac{9}{4}e^\nu A \in [0.01821610682118158624145957712583, 0.01821610682118158624145957712603],$$

whose six-decimal rounding is 0.018216. $\square$

Corroboration, not part of the proof. From the exact b-file values, $g_n/(0.018216 n! \rho^{-n} n^{-5/2}) = 0.48866, 0.46548, 0.45472, 0.45055$ at $n = 30, 60, 120, 200$, decreasing towards $e^\nu A/0.018216 = 0.44444\dots$, while the relative residual of g_n after the four corrections of (3.10) is $1.37 \cdot 10^{-5}, 4.13 \cdot 10^{-7}, 1.27 \cdot 10^{-8}, 9.81 \cdot 10^{-10}$ (and of c_n after those of (3.9): $5.69 \cdot 10^{-6}, 1.71 \cdot 10^{-7}, 5.27 \cdot 10^{-9}, 4.07 \cdot 10^{-10}$). The introduction of the same article (pp. 2092–2093) prints $h_n \sim h \cdot n^{-3/2} \sigma^{-n} n!$ with $\sigma \approx 0.13659$, a shifted Poisson parameter $\xi = 0.14840$ and the connectivity limit $e^{-\xi} = 0.86208$. These are wrong as well: by (i), $g_n n^{3/2} \rho^n/n! \sim e^\nu A/n \rightarrow 0$, so no positive h satisfies the introduction's formula; and Theorem 6.1 with $\lambda_n \rightarrow \nu$ gives the limit law $1 + \text{Poisson}(\nu)$, $\nu = 0.1488867426\dots$, so 0.14840 is wrong in its fourth decimal and 0.86208 (which is $e^{-0.14840} = 0.862086\dots$ truncated) differs from $e^{-\nu} = 0.8616666\dots$. BGKN's own Theorem 6.1 (p. 2103) has the right values $\xi \approx 0.14889$ and $e^{-\xi} \approx 0.86166$, and its table (p. 2105) 0.1489 and 0.8616. Section 9 states all of this; the write has read the pages and supplied the proof. No cause is established (question 5 of Section 10).

Remark 9.2 ([write] Other versions and BGKN's Theorem 7.1, 6 October 2026). (a) *The*

preprint [5] (version 1, 19 December 2005, the only version listed). Its Theorem 5.1 has the same form with “ $\rho \approx 0.13659$ and $h \approx 0.017657$ ”: wrong by the proof of Remark 9.1, being 2.18094... times $e^\nu A$ (not $\frac{9}{4}$). Its Theorem 6.1 prints the parameter 0.14840 and the limit 0.86208, the values that the published introduction retains, and its introduction has the exponent $-3/2$. Its Theorem 7.1 prints $s \approx 0.013961$ (item (d)).

(b) *The EuroComb 2005 extended abstract* [11]: its Theorem 3 gives h_n with exponent $-3/2$ and the radius 0.136593, and no value of the amplitude.

(c) *Kang's thesis* [6]. Section 5.2 (p. 55) gives $C(R) \doteq 0.148886$ and $\alpha_1 \doteq 0.006976$ (truncations of ν and A), the relation $\alpha_0 = \alpha_1 e^{C(R)}$ for the all-graph amplitude, and “ $\alpha_0 = \alpha_1 e^{C(R)} \doteq 0.008095$ ”; Theorem 5.2.1 (stated on p. 9) prints $c_1 \doteq 0.008095$ and $\gamma \doteq 7.32098$, and Theorem 5.3.2 the connectivity limit 0.861666 (a truncation of $e^{-\nu}$). So, as

Section 9 says, the thesis has the correct analytic normalization, and its decimal 0.008095 is neither the rounding nor the truncation of $e^\nu A = 0.0080960\dots$ (both are 0.008096). The product of its two printed values, $0.006976 e^{0.148886} = 0.0080959318\dots$, truncates to 0.008095; whether the printed value arose that way is not known.

- (d) *BGKN's Theorem 7.1* (graphs with no $K_{2,3}$ minor, p. 2104) reads “ $s_n \sim s \cdot n^{-5/2} \cdot \rho^{-n} n!$ where $\rho \approx 0.13648$ and $s \approx 0.018288$ ”. By BGKN's own description its block function is $b(u) + u^3/6$ (the outerplanar blocks and the single labelled K_4). The write evaluated the same constants for this block function: the saddle $0.1707605103918\dots$, the radius $0.1364804209511\dots$, $\nu' = 0.1487804550899\dots$ and the amplitude $e^{\nu'} A' = 0.0081282385507081\dots$. Then $0.018288 = 2.24993\dots \times e^{\nu'} A'$: it is $\frac{9}{4} e^{\nu'} A' = 0.0182885367\dots$ truncated to six decimals, the same factor as in Theorem 5.1, while its Theorem 7.3 values 0.14879 and 0.86175 agree with ν' and $e^{-\nu'} = 0.861758\dots$ to within one unit of their last digits. Exact counts of this class from (2.4) with $b + u^3/6$ (the write's own code; $s_1, \dots, s_6 = 1, 2, 8, 64, 918, 20284$) give $s_n / (e^{\nu'} A' n! \rho'^{-n} n^{-5/2}) = 1.0735, 1.0354, 1.0175$ at $n = 40, 80, 159$. This is an observation, not a theorem of this report: Theorem 3.1 is stated for b only (question 6).

10 Further questions

Several next steps would add information beyond merely printing more terms from the same generator.

1. Evaluate explicit remainder constants and starting thresholds, then combine them with interval arithmetic to turn Theorem 7.1 into a finite-input certified algorithm.
2. Determine the growth of $d_j(v)$ as j increases. The fixed-order proof does not address optimal truncation, divergence, or exponentially small corrections.
3. Extract higher-order signed corrections to the shifted-Poisson law and quantify their total-variation effect. Positivity of a proposed finite-order distributional approximation requires care.
4. Extend the marked exact cancellation to edge statistics jointly with components, retaining complex uniformity and the abstract-graph normalization. The leading edge-count limit theory is already classical.

[write] **Added 6 October 2026, under Vladimir's standing rule of 4 October 2026.** The source's four questions stand; Remark 7.2 makes the monotonicity onset of question 1 explicit ($n_1 = 1$ for g , 2 for c), leaving the remainder constants and the error threshold. Three questions are added; the first two come from Section 9 and Remark 9.2, the third from the non-claim of Section 5.1.

5. *The cause of BGKN's factor.* Find the step that makes BGKN's printed amplitudes $\frac{9}{4}$ times the true ones (Remark 9.1; the source: “No cause of that discrepancy is established here”), and decide whether BGKN's series-parallel constants $c \approx 0.0067912$ and $g \approx 0.0076388$ (its Theorem 3.7, computed by $c = 3C_3/(4\sqrt{\pi})$, $g = \exp(C_0)c$) are affected. The recurrence of the same factor in its Theorem 7.1 (Remark 9.2(d)) suggests a slip common to the two outerplanar-type computations; the write has not located it. Missing: a recomputation of the series-parallel constants from BGKN's implicit equations, and the preprint's different value 0.017657 explained.

6. *Graphs with no $K_{2,3}$ minor.* Prove Theorem 3.1 for the block function $b(u) + u^3/6$ and so turn Remark 9.2(d) into a correction of BGKN's Theorem 7.1 ($s = e^{\nu'} A' = 0.0081282\dots$, not 0.018288). Sketch: the proofs of Sections 2–4 use of b only that its coefficients are nonnegative, that $b_1 = 1$, that $ub'(u)$ increases to infinity on $(0, R)$ and that the saddle lies inside the disc of analyticity, and $u^3/6$ preserves all four. Missing: the restatement and the check that no step uses the special form of (2.1) (the field $\mathbb{Q}(\tau)$ and the quartic (3.2) change).
7. *A second analytic route.* Promote the formal singular expansion of Section 5.1 to a proof: the source says one “must supply suitable Δ -domain continuation and transfer remainder hypotheses, as in the established subcritical theory”, and that “that additional route is not silently assumed here”. Sketch: the subcritical framework [7] applied to $T = ze^{b(T)}$. Missing: the continuation of T to a Δ -domain and the transfer of each fixed order, uniformly in the marker.

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